Interactive Prototype Evaluation 3 and Reflection

1 Prototype Session Review

1.1 What Worked

The strongest features make action results visible and connect them to the current activity. These strengths concern feedback after an action, rather than independent discovery of the whole task. (see Appendix A1).

Ancestor interaction

All five desktop cases record a three-segment energy increase after using the reward, with understanding rated 4.60/5. The change affects a resource needed for flight, explaining both what happened and why the reward matters. This follows IP2a, where appearance changes confirmed mask equipment. These results support visible state feedback, although the reward task does not show whether players can independently decide when to restore energy. (see Appendix A1; F2; F3).

Flight interaction

Flame scoring receives 4.40/5. P01 collects three flames for 30 points and P04 collects four for 40 points. Immediate updates and a consistent ten points per flame make the rule easy to follow. P03 understands scoring despite stopping, showing that clear feedback and comfortable movement need separate evaluation. These records support the scoring explanation, but do not establish sustained enjoyment or headset comfort. (see Appendix A1; A2; A4; A5).

Communication

All five cases send and read back the typed phrase; four complete typing independently, with a rating of 4.40/5. A visible draft and posted message allow checking before and after sending. However, physical-keyboard familiarity may contribute to this result. The phrase is only "we can fly", so this supports basic desktop entry and display rather than longer conversation or controller typing. (see Appendix A1; A2; A3; A7).

1.2 What Did Not Work Well

The main difficulties occur before an action or while continuing a task. Completion needs to be read alongside assistance, comfort and the distinction between typed and full communication outcomes. (see Appendix A0; A1).

Ancestor interaction

Only three cases complete the reward task independently, below the four-out-of-five target. P02 notices the ancestor at four seconds but obtains the reward at 42 seconds after three prompts. Attention therefore does not explain how to open the exchange; greater brightness alone may have limited value. P04 also presses Q when intending to put an item down. This suggests insufficient information before choosing an action.

Flight interaction

Three cases complete flight independently, one needs assistance and one stops. P02 needs the take-off control explained. P03 stops during forward movement at 34 seconds as discomfort rises from 0 to 5, despite understanding scoring. P05 pauses to reorient after switching views. These events concern different stages: P03 had not switched views, so these events do not show that they share a cause. The records concern monitor use and cannot identify which headset movement setting would be comfortable. (see Appendix A1; A3; A4; A6; A7).

Communication

P02 finishes the draft at 27 seconds but sends at 59 seconds after help because Enter is unclear. The obstacle is sending rather than entry. Q15 averages 3.60/5 across instructions, resource displays and chat; it is not a dedicated chat-text score. P01 and P05 provide the specific evidence about small chat text. Together, these examples distinguish action guidance from the effort needed to inspect a result. (see Appendix A1; A2; A3; A6; G1).

The separate speech scenarios introduce 2.9-3.2-second waits and three incorrect first outputs. P02 and P05 correct the result after retrying; P03 still receives changed meaning. Small captions can also delay checking. These examples explain why processing feedback, correction and readability need separate attention.

1.3 Overall Summary

The desktop records show clear reward and scoring feedback but only three independent completions for both reward use and flight. Full communication remains blocked, although typing is separately interpretable. I would retain visible feedback while treating discovery, comfort and complete communication as unresolved design concerns. These conclusions remain specific to the record type and platform. (see Appendix A0; A1; B2; C1).

2 Methodological Reflection

I used low-fidelity models, task testing, observation, questionnaires and interviews across the four stages. IP1 used Unity keyboard control; IP2a used Meta Quest. IP2b had a Quest protocol and a separate desktop adaptation. Keeping those records distinct is essential when assessing what each method can explain. (see Appendix D2; E1; F1; B1; A7).

2.1 Appropriateness of the Methods

Design Concept

Screenshots, sketches and paper models let me change menu positions and action sequences quickly. The overhead-menu model made the intended reach and possible view obstruction concrete. However, a paper action does not reproduce tracking delay, headset viewing limits or fatigue from repetition. The method suited early layout questions, but questions about repeated reaching required more than a single demonstration. (see Appendix D1; D2).

IP1

Unity keyboard testing made movement, camera behaviour and object functions easy to observe and change. It also avoided teaching headset controls during an early functional check. However, I explained E, Q, P and T in Interview 1 and introduced the keyboard verbally. Later completion therefore included learning from my instructions. Recording the initial attempt separately from guided performance would better show discoverability; keyboard errors also needed to remain separate from VR-gesture difficulties. (see Appendix E1; E2; E3).

IP2a

The headset tasks matched the ancestor and mask design more closely. Separate limits of 40, 60 and 50 seconds helped locate difficulties in opening, obtaining and wearing. However, preparing the next state after a failure could hide a broken overall sequence. I needed to report both individual tasks and the complete chain, including any state I prepared. The plan's neutral practice also meant that independent completion referred to the task period after basic input training. (see Appendix F1).

IP2b

The Quest plan connected rewards, flight and communication, with substeps and explicit assistance rules. This structure suited locating a pause or incorrect action. However, three tasks plus 15 questions increased recording and recall demands. A 90-second flight round also makes total task time a poor efficiency measure. I would separate waiting, active input and technical delay, with short follow-up questions after demanding actions. Desktop timings would stay outside any headset-performance comparison. (see Appendix B1; B2; G1; A7).

2.2 Effectiveness in Producing Actionable Insights

Design Concept

Early exploration identified situations where voice alone would be unsuitable, including quiet settings. This supported retaining a silent input option. However, the models compared possible use situations rather than speed or accuracy. They justified multimodal choice, but choosing a finger-worn device over another text method required a comparison using the same message and setting. (see Appendix D1; D2).

IP1

A tester with two years of Sky experience still needed the pickup key explained and repeatedly adjusted the camera. This separated game familiarity from understanding the new controls. However, inconsistent help and error records made it difficult to rank problems. Recording the action, interruption and help level would identify whether the main obstacle was discovering the control or executing it. (see Appendix E2; E3).

IP2a

Purpose clarity scored 4.6/5, while discovery scored 4.0/5. Three participants were unsure whether the mask was selected or obtained; P01 expected it in the hand and P05 wanted a status message. These details pointed to acquisition feedback and item location. Satisfaction of 4.8/5 described the overall impression, but gave less guidance about which step to change. (see Appendix F2; F3).

IP2b

Separating access, entry, sending and readback made the communication analysis more useful. A completed draft that is not sent needs clearer action guidance; a posted message that is hard to read needs a display change. Calling both "poor communication" would provide little design direction. However, Q15 combines several displays, so the specific event and follow-up comment are needed to locate the problem.(see Appendix B1; G1; A3; A6; C1).

2.3 Overall Summary

The methods were most useful when they connected a specific event with an explanation. Paper models addressed layout; Unity testing exposed control and camera issues; the focused mask test identified state uncertainty. Clearer help records, a whole-sequence outcome and narrower questions would strengthen the design decisions. Platform and data status must remain visible when combining the stages. (see Appendix D2; E3; F1; F3; B2).

3 Concept Evaluation

I consider the original XR concept partially supported. IP1 and IP2a support interest in direct object interaction and the value of visible feedback. This makes the support strongest for understandable object outcomes and more limited for comfortable flight and smooth communication. (see Appendix D1; E2; F2; F3; A0; C1).

In IP2a, mask-wearing scored 4.6/5 and appearance changes confirmed success, while three participants remained unsure about acquisition. A spatial object therefore still needs clear states. The IP2b flight examples also show that understanding scoring can coexist with discomfort or reorientation. I learned to treat understandable feedback and comfortable control as separate requirements. (see Appendix F2; F3; A4; A6).

The concept should retain text and speech as choices. Drafts allow checking, while the speech scenarios explain possible waiting and correction costs. Sending a fixed phrase gives limited evidence about natural conversation or emotional connection. I would refine the concept into a VR experience with comfortable exploration, clear item states, controllable flight and communication that players can check and correct. (see Appendix D1; A3; C1; C2).

4 Improvements and Extensions

I would retain visible mask equipment, energy changes, throwing previews and scoring feedback. The five Week 6 principles help apply the findings to the points where players discover, control and check actions. (see Appendix F3; A1; A2; D3).

For text readability, I would provide adjustable text sizes, stronger background separation and a stable chat panel. P01 and P05 identify the reading effort, while Q15 only supplies a broader readability measure. Speech results should appear as editable drafts with clear processing and sending states, allowing waiting and incorrect output to be recognised separately. (see Appendix A2; A6; G1; C2; D3).

For ergonomics, common menus should sit within comfortable reach in front of the body. Arm-spreading could remain optional alongside smaller movements or buttons, and the keyboard should support seated positioning. These are changes to the original proposals for movement and overhead menus; the low-fidelity models do not establish long-term physical comfort. (see Appendix D1; D2; D3).

For brightness changes, I would reduce sudden large flashes during collection and use a local glow, score update or light vibration. P04's example explains the risk of distracting attention from direction. Ancestor highlighting also needs an action prompt, since P02 notices the character before understanding the exchange control. (see Appendix A5; A3; D3).

For avoiding simulator sickness, I would offer slower movement, less sudden acceleration and an accessible stop control. View changes should be player-initiated and keep direction consistent where possible. For button placement, prompts should sit near relevant objects, with distinct drop, use and throw actions, a cancellable preview and a fixed send control. These proposals respond to the recorded difficulties while keeping desktop controls and discomfort separate from Quest findings. (see Appendix A3; A4; A5; A6; B2; D3).

MR could extend stationary tea-table communication and object inspection near a real table, with seated use and less dependence on continuous virtual travel. Readability and recognition would still need separate fixes. VR remains better aligned with the concept's continuous open sky; a real room would constrain that flight experience. Mobile AR could introduce masks and reward effects before VR. These are concept extensions, not tested improvements. (see Appendix D1; D3; B2).

Appendix A IP2b Desktop Case Records

A0 Record status and outcome definitions

I = independent completion; A = completion after H1/H2 help; P = partial completion; F = unsuccessful attempt; X = technical block; S = stopped for comfort; N = unattempted. H0 repeats the goal, H1 directs attention to a cue, and H2 supplies the exact control. Independent completion allows logged H0 but no H1/H2. The original voice steps remain N, full Task 3 remains X, and Q14 remains N/A.

A1 Task outcomes and questionnaire summary

Case	Reward task	Flight task	Full communication	Typing
P01	I / 66 s	I / 108 s	X / 50 s	I
P02	A / 88 s	A / 116 s	X / 85 s	A
P03	I / 74 s	S / 34 s	X / 55 s	I
P04	A / 85 s	I / 106 s	X / 62 s	I
P05	I / 79 s	I / 112 s	X / 65 s	I

Measure	P01	P02	P03	P04	P05	Mean	n
Q1 Notice ancestor	5	4	4	5	4	4.40	5
Q2 Open exchange	4	2	4	5	4	3.80	5
Q3 Obtain reward	5	3	4	5	4	4.20	5
Q4 Drop versus throw	5	3	4	2	4	3.60	5
Q5 Aim preview	5	4	5	4	4	4.40	5
Q6 Consumption	5	3	4	4	4	4.00	5
Q7 Energy change	5	4	5	5	4	4.60	5
Q8 Take-off/ascent	5	2	4	5	4	4.00	5
Q9 Energy display	4	3	3	5	3	3.60	5
Q10 Flame scoring	5	4	4	5	4	4.40	5
Q11 End refill	5	3	N/A	5	4	4.25	4
Q12 Orientation	4	3	N/A	4	3	3.50	4
Q13 Type and send	5	3	5	5	4	4.40	5
Q14 Voice	N/A	N/A	N/A	N/A	N/A	N/A	0
Q15 Combined readability	4	3	4	4	3	3.60	5

Ratings run from 1 (strongly disagree) to 5 (strongly agree); N/A is excluded from the mean. Q11 and Q12 each have four valid cases; Q14 has none. Q15 covers instructions, resource displays and chat together. All five reward examples show +3 energy; four pottery throws reach the target. The local task target is four independent completions out of five, not a statistical threshold.

A2 P01 observation record

Assigned order: Typed first. Actual route: typed only after the configuration gate. No microphone attempt. Initial energy 10, wallet 50; prepared start state as above. Monitor/device/build details not measured.

Substep	Code	Elapsed s	H0/H1/H2
T1a Appearance	I	5	0/0/0
T1b Acquire	I	17	0/0/0
T1c Drop and pickup	I	31	0/0/0
T1d Consume	I	44	0/0/0
T1e Throw	I	66	0/0/0
T2a Start round	I	8	0/0/0
T2b Take-off and ascent	I	16	0/0/0
T2c Flame	I	27	0/0/0
T2d View switch	I	46	0/0/0
T2e Natural refill	I	108	0/0/0
T3a Table access	I	12	0/0/0
T3b Typed message	I	34	0/0/0
T3c Voice gated	N	—	N/A
T3d Readback incomplete	P	38	0/0/0

Whole tasks: T1 I 66 s; T2 I 108 s; T3 X 50 s. Full T3 blocked; its typed substep remains separately interpretable.

Assistance log

Explicitly no H0/H1/H2 assistance in any task. No blank help fields are interpreted as zero.

T1 energy 10→13. T2 take-off 15→14; additional ascent 14→13. Flames 3; score 30. Natural final energy 15. Discomfort baseline / pre-flight / post-flight / final: 0 / 0 / 1 / 0. Local 0–10 screen-use ratings only.

A2 P01 ratings and task trace

Questionnaire statement	Rating
01. I noticed the ancestor when I approached the monument.	5
02. I could work out how to open the exchange tree.	4
03. I could obtain the reward I intended to select.	5
04. I could distinguish dropping an item from throwing it.	5
05. The throwing arc and landing circle helped me aim.	5
06. I understood how to consume the energy reward.	5
07. I understood the resource change after consuming the reward.	5
08. I understood how to take off and boost.	5
09. I could tell how much flight energy remained.	4
10. I understood how collecting flames affected my score.	5
11. I understood why my energy refilled when the round ended.	5
12. I remained oriented when switching views.	4
13. I could enter and send the typed message.	5
14. I could send the intended message using my voice.	N/A
15. I could read the instructions, resource displays and chat bubbles.	4

Scale 1 strongly disagree to 5 strongly agree. Q8 “boost” means desktop ascent in this adaptation. Q14 N/A: no voice attempt. Preferred view: Third person. Input comparison: unavailable, only typing experienced.

Chronological task trace

T1: ancestor noticed 5 s; F and number/Enter acquisition 17 s; Z drop/pickup 31 s; E consumption 44 s; Q preview/release 66 s. Pottery lands in the target area. Energy 10→13.

T2: start 8 s, energy refill to 15; take-off 13 s changes 15→14; one further ascent pulse changes 14→13 by 16 s. First flame 27 s; 3-key view switch 46 s. Three flames; 30 points. Natural timeout 98 s; explanation complete 108 s.

T3: reward-table acquisition and Q placement, then table click, complete 12 s. Physical typing starts 17 s; matching message sent with Enter 34 s and read back 38 s. One wrong key is corrected before send. Voice is not attempted because the preflight configuration is unavailable; final record closes 50 s.

Interview responses

Which moment was most confusing, and what did you expect? “When I aimed the pottery, the arc showed me roughly where it would land before I released Q. Reading the chat bubble took more effort. After I sent the message, I had to look closely to check that it showed what I had typed.”

Which single change would help you most? “I would enlarge just the text in the chat bubble. After pressing Enter, I want to check the whole sentence while keeping the table and the rest of the scene in view.”

Why did you prefer that view or input method? “I preferred third person for flight because I could see which way the character was facing before I turned. I only tried the keyboard at the table, so I can comment on reading the text but cannot compare it with speech.”

A3 P02 observation record

Assigned order: Voice first. Actual route: typed only after the configuration gate. No microphone attempt. Initial energy 11, wallet 50; prepared start state as above. Monitor/device/build details not measured.

Substep	Code	Elapsed s	H0/H1/H2
T1a Appearance	I	4	0/0/0
T1b Acquire	A	42	1/1/1
T1c Drop and pickup	I	56	0/0/0
T1d Consume	I	70	0/0/0
T1e Throw	I	88	0/0/0
T2a Start round	I	8	0/0/0
T2b Take-off and ascent	A	43	1/1/1
T2c Flame	I	60	0/0/0
T2d View switch	I	75	0/0/0
T2e Natural refill	I	116	0/0/0
T3a Table access	I	15	0/0/0
T3b Typed message	A	59	1/1/0
T3c Voice gated	N	—	N/A
T3d Readback incomplete	P	67	0/0/0

Whole tasks: T1 A 88 s; T2 A 116 s; T3 X 85 s. Full T3 blocked; its typed substep remains separately interpretable.

Assistance log

Task / second	Level	Exact wording
T1b / 14	H0	What would you try next?
T1b / 24	H1	Look at the ancestor exchange control cue.
T1b / 34	H2	Press F near the ancestor, select the energy reward with its number key, then press Enter.
T2b / 18	H0	What would you try next?
T2b / 28	H1	Look at the take-off instruction.
T2b / 38	H2	Press Space to take off; a further ascent input can spend another segment.
T3b / 37	H0	What would you try next?
T3b / 47	H1	Look at the send instruction beside the draft.

T1 energy 11→14. T2 take-off 15→14; additional ascent 14→13. Flames 1; score 10. Natural final energy 15. Discomfort baseline / pre-flight / post-flight / final: 0 / 0 / 2 / 1. Local 0–10 screen-use ratings only.

A3 P02 ratings and task trace

Questionnaire statement	Rating
01. I noticed the ancestor when I approached the monument.	4
02. I could work out how to open the exchange tree.	2
03. I could obtain the reward I intended to select.	3
04. I could distinguish dropping an item from throwing it.	3
05. The throwing arc and landing circle helped me aim.	4
06. I understood how to consume the energy reward.	3
07. I understood the resource change after consuming the reward.	4
08. I understood how to take off and boost.	2
09. I could tell how much flight energy remained.	3
10. I understood how collecting flames affected my score.	4
11. I understood why my energy refilled when the round ended.	3
12. I remained oriented when switching views.	3
13. I could enter and send the typed message.	3
14. I could send the intended message using my voice.	N/A
15. I could read the instructions, resource displays and chat bubbles.	3

Scale 1 strongly disagree to 5 strongly agree. Q8 “boost” means desktop ascent in this adaptation. Q14 N/A: no voice attempt. Preferred view: First person. Input comparison: unavailable, only typing experienced.

Chronological task trace

T1: notices ancestor 4 s but cannot open exchange. Prompts at 14/24/34 s; reward acquired 42 s, Z drop/pickup 56 s, E consumption 70 s and Q throw 88 s. Pottery lands outside target. Energy 11→14. One wrong exchange key before assistance.

T2: start 8 s and refill 15; prompts occur while grounded. Take-off after H2; recorded 15→14, then a further ascent pulse 14→13 by 43 s. First flame 60 s; view switch 75 s. One flame, 10 points; timeout 98 s, explanation 116 s.

T3: acquired table, placed with Q and clicked it by 15 s. Physical typing starts 20 s and target draft is ready by 27 s; participant hesitates about sending. H0 37 s and H1 47 s; Enter sends at 59 s, readback 67 s. Voice-first assignment cannot run; actual order is typed only after the preflight gate. Record closes 85 s.

Interview responses

Which moment was most confusing, and what did you expect? “I saw the ancestor and knew I needed an energy reward, but standing beside it did not tell me that F opened the exchange. Later I had the whole sentence in the draft and stopped again. I was unsure whether Enter would send it or start a new line.”

Which single change would help you most? “I would put the key beside the action I can take right now. I needed “F to open exchange” beside the ancestor and “Enter to send” beside my finished draft. Those were the two places where I waited for help.”

Why did you prefer that view or input method? “First person made my flight direction easier to follow. With typing, I could see that my draft was ready, so I knew that sending was the part I was stuck on. I did not try voice and cannot say whether it would be easier.”

A4 P03 observation record

Assigned order: Typed first. Actual route: typed only after the configuration gate. No microphone attempt. Initial energy 12, wallet 50; prepared start state as above. Monitor/device/build details not measured.

Substep	Code	Elapsed s	H0/H1/H2
T1a Appearance	I	6	0/0/0
T1b Acquire	I	22	0/0/0
T1c Drop and pickup	I	38	0/0/0
T1d Consume	I	56	0/0/0
T1e Throw	I	74	0/0/0
T2a Start round	I	7	0/0/0
T2b Take-off and ascent	I	16	0/0/0
T2c Flame	I	24	0/0/0
T2d View switch	N	—	N/A
T2e Natural refill	N	—	N/A
T3a Table access	I	14	0/0/0
T3b Typed message	I	39	0/0/0
T3c Voice gated	N	—	N/A
T3d Readback incomplete	P	44	0/0/0

Whole tasks: T1 I 74 s; T2 S 34 s; T3 X 55 s. Full T3 blocked; its typed substep remains separately interpretable.

Assistance log

Explicitly no H0/H1/H2 assistance in any task. No blank help fields are interpreted as zero.

T1 energy 12→15. T2 take-off 15→14; additional ascent 14→13. Flames 1; score 10. Natural final energy not observed. Discomfort baseline / pre-flight / post-flight / final: 0 / 0 / 5 / 1. Local 0–10 screen-use ratings only.

Safety action at 34 s: “We will stop now.” Screen task paused immediately. Five-minute seated break excluded from task timing. The case resumes at the stationary table after the break.

A4 P03 ratings and task trace

Questionnaire statement	Rating
01. I noticed the ancestor when I approached the monument.	4
02. I could work out how to open the exchange tree.	4
03. I could obtain the reward I intended to select.	4
04. I could distinguish dropping an item from throwing it.	4
05. The throwing arc and landing circle helped me aim.	5
06. I understood how to consume the energy reward.	4
07. I understood the resource change after consuming the reward.	5
08. I understood how to take off and boost.	4
09. I could tell how much flight energy remained.	3
10. I understood how collecting flames affected my score.	4
11. I understood why my energy refilled when the round ended.	N/A
12. I remained oriented when switching views.	N/A
13. I could enter and send the typed message.	5
14. I could send the intended message using my voice.	N/A
15. I could read the instructions, resource displays and chat bubbles.	4

Scale 1 strongly disagree to 5 strongly agree. Q8 “boost” means desktop ascent in this adaptation. Q14 N/A: no voice attempt. Preferred view: Not both experienced. Input comparison: unavailable, only typing experienced.

Chronological task trace

T1: ancestor 6 s, acquisition 22 s, drop/pickup 38 s, E consumption 56 s, Q throw 74 s. Pottery lands in target. Energy 12→15. No help.

T2: start 7 s/refill 15; take-off 12 s changes 15→14, further ascent pulse 14→13 by 16 s; first flame 24 s. At 34 s participant requests a stop during forward screen motion. No view switch or natural refill observed. One flame, 10 points.

T3: after a five-minute break, the case specifies symptoms eased to 1/10 and voluntary continuation with a stationary screen task. Table access 14 s; typing starts 20 s, send 39 s, readback 44 s. Voice not attempted due to preflight configuration gate. Record closes 55 s.

Interview responses

Which moment was most confusing, and what did you expect? “I had collected one flame and understood the score change. When the scene kept moving forwards, I began to feel uncomfortable. I wanted the movement to stop before trying another view or waiting for the round to end.”

Which single change would help you most? “I would keep a pause control visible throughout flight. Once I felt uncomfortable, I wanted to stop immediately without watching the scene move while searching for a key. A gentler camera movement would also help.”

Why did you prefer that view or input method? “I stopped before switching views, so I cannot pick a more comfortable view. After the break, the stationary scene at the table let me focus on typing the message and reading its bubble.”

A5 P04 observation record

Assigned order: Voice first. Actual route: typed only after the configuration gate. No microphone attempt. Initial energy 9, wallet 50; prepared start state as above. Monitor/device/build details not measured.

Substep	Code	Elapsed s	H0/H1/H2
T1a Appearance	I	4	0/0/0
T1b Acquire	I	8	0/0/0
T1c Drop and pickup	A	50	1/1/0
T1d Consume	I	65	0/0/0
T1e Throw	I	85	0/0/0
T2a Start round	I	6	0/0/0
T2b Take-off and ascent	I	15	0/0/0
T2c Flame	I	22	0/0/0
T2d View switch	I	40	0/0/0
T2e Natural refill	I	106	0/0/0
T3a Table access	I	12	0/0/0
T3b Typed message	I	36	0/0/0
T3c Voice gated	N	—	N/A
T3d Readback incomplete	P	42	0/0/0

Whole tasks: T1 A 85 s; T2 I 106 s; T3 X 62 s. Full T3 blocked; its typed substep remains separately interpretable.

Assistance log

Task / second	Level	Exact wording
T1c / 28	H0	What would you try next?
T1c / 38	H1	Look at the separate pickup/drop and aim/launch key labels.

T1 energy 9→12. T2 take-off 15→14; additional ascent 14→13. Flames 4; score 40. Natural final energy 15. Discomfort baseline / pre-flight / post-flight / final: 0 / 0 / 1 / 0. Local 0–10 screen-use ratings only.

A5 P04 ratings and task trace

Questionnaire statement	Rating
01. I noticed the ancestor when I approached the monument.	5
02. I could work out how to open the exchange tree.	5
03. I could obtain the reward I intended to select.	5
04. I could distinguish dropping an item from throwing it.	2
05. The throwing arc and landing circle helped me aim.	4
06. I understood how to consume the energy reward.	4
07. I understood the resource change after consuming the reward.	5
08. I understood how to take off and boost.	5
09. I could tell how much flight energy remained.	5
10. I understood how collecting flames affected my score.	5
11. I understood why my energy refilled when the round ended.	5
12. I remained oriented when switching views.	4
13. I could enter and send the typed message.	5
14. I could send the intended message using my voice.	N/A
15. I could read the instructions, resource displays and chat bubbles.	4

Scale 1 strongly disagree to 5 strongly agree. Q8 “boost” means desktop ascent in this adaptation. Q14 N/A: no voice attempt. Preferred view: Third person. Input comparison: unavailable, only typing experienced.

Chronological task trace

T1: ancestor 4 s; reward acquired 8 s. Participant uses Q instead of Z and unintentionally launches reward at 16 s; no progress from 18 s. H0 28 s, H1 38 s. Recovery and deliberate Z drop/pickup 50 s, E consumption 65 s, pottery throw 85 s. Pottery lands in target. Energy 9→12.

T2: start 6 s/refill 15; take-off 11 s changes 15→14 and further ascent pulse 14→13 by 15 s. First flame 22 s, view switch 40 s. Four flames, 40 points; timeout 96 s and explanation 106 s. Brief bright collection effect described as distracting.

T3: planned voice-first; actual typed-only after voice gate. Table access 12 s, typing starts 18 s, sends 36 s, readback 42 s. While typing the participant reports a small viewpoint/movement shift; no causal diagnosis is claimed. Voice not attempted. Record closes 62 s.

Interview responses

Which moment was most confusing, and what did you expect? “I wanted to set the reward down at my feet. I pressed Q and it flew away. After the prompt I understood that Z put it down and Q aimed and launched it. Getting the reward had been straightforward, so its sudden launch surprised me.”

Which single change would help you most? “I would show “Z to drop” and “Q to aim and throw” together while I am carrying something. I need that distinction when I decide what to do with the item. By the time the throwing arc appears, I have already pressed the wrong key.”

Why did you prefer that view or input method? “I preferred third person in flight because I could see the character’s heading. When I typed at the table the view seemed to shift a little. I could not tell why, but I wanted it to stay still while I checked the message.”

A6 P05 observation record

Assigned order: Typed first. Actual route: typed only after the configuration gate. No microphone attempt. Initial energy 10, wallet 50; prepared start state as above. Monitor/device/build details not measured.

Substep	Code	Elapsed s	H0/H1/H2
T1a Appearance	I	5	0/0/0
T1b Acquire	I	23	0/0/0
T1c Drop and pickup	I	41	0/0/0
T1d Consume	I	60	0/0/0
T1e Throw	I	79	0/0/0
T2a Start round	I	9	0/0/0
T2b Take-off and ascent	I	18	0/0/0
T2c Flame	I	31	0/0/0
T2d View switch	I	52	0/0/0
T2e Natural refill	I	112	0/0/0
T3a Table access	I	13	0/0/0
T3b Typed message	I	42	0/0/0
T3c Voice gated	N	—	N/A
T3d Readback incomplete	P	49	0/0/0

Whole tasks: T1 I 79 s; T2 I 112 s; T3 X 65 s. Full T3 blocked; its typed substep remains separately interpretable.

Assistance log

Explicitly no H0/H1/H2 assistance in any task. No blank help fields are interpreted as zero.

T1 energy 10→13. T2 take-off 15→14; additional ascent 14→13. Flames 2; score 20. Natural final energy 15. Discomfort baseline / pre-flight / post-flight / final: 0 / 0 / 2 / 1. Local 0–10 screen-use ratings only.

A6 P05 ratings and task trace

Questionnaire statement	Rating
01. I noticed the ancestor when I approached the monument.	4
02. I could work out how to open the exchange tree.	4
03. I could obtain the reward I intended to select.	4
04. I could distinguish dropping an item from throwing it.	4
05. The throwing arc and landing circle helped me aim.	4
06. I understood how to consume the energy reward.	4
07. I understood the resource change after consuming the reward.	4
08. I understood how to take off and boost.	4
09. I could tell how much flight energy remained.	3
10. I understood how collecting flames affected my score.	4
11. I understood why my energy refilled when the round ended.	4
12. I remained oriented when switching views.	3
13. I could enter and send the typed message.	4
14. I could send the intended message using my voice.	N/A
15. I could read the instructions, resource displays and chat bubbles.	3

Scale 1 strongly disagree to 5 strongly agree. Q8 “boost” means desktop ascent in this adaptation. Q14 N/A: no voice attempt. Preferred view: First person. Input comparison: unavailable, only typing experienced.

Chronological task trace

T1: ancestor 5 s, reward 23 s, drop/pickup 41 s, E consumption 60 s, Q throw 79 s. Pottery lands in target. Energy 10→13. Brief search for consumption cue, under the assistance interval.

T2: start 9 s/refill 15; take-off 14 s changes 15→14 and further ascent pulse 14→13 by 18 s. First flame 31 s, view switch 52 s. Pauses to reorient, returns to first person. Two flames, 20 points; timeout 99 s and explanation 112 s.

T3: table access 13 s; typing starts 18 s, sends 42 s after correcting one wrong key; reads bubble at 49 s. Reports small text on the screen. Voice not attempted after configuration gate; closes 65 s.

Interview responses

Which moment was most confusing, and what did you expect? “After switching views, I paused to work out which way I had been flying. At the table I corrected a wrong key before sending. The text looked small enough that I had to look closely to check the bubble against what I had typed.”

Which single change would help you most? “If I could change one thing, I would keep a heading cue visible through the view change. A brief indication of which way the character is facing would help me continue along the same route without pausing to find my direction.”

Why did you prefer that view or input method? “I tried both views and went back to first person because I could better predict which way I would move. I only used the keyboard, so I can describe correcting and sending my text but cannot compare that experience with voice.”

A7 Desktop task conditions and input routes

Action	Verified desktop input path
Movement and view	WASD and mouse; 1 first person / 3 third person when exchange is closed.
Exchange	F near ancestor; number keys select and Enter acquires. Mouse drag acquisition also exists. Record the route actually used.
Reward handling	Z drops/picks up; E consumes the edible reward; hold Q shows aim and release launches. No controller-to-face task.
Flight	Space initiates flight; continuing ascent may consume another segment after 0.8 s. Left Ctrl descends. Record take-off cost separately from later ascent costs.
Tea table	For the plan-aligned route, acquire the table reward and use Q placement, then click it. P preview / R placement is an alternative and must be labelled as a deviation.
Typed chat	Physical typing with Enter or on-screen SEND; table click or T/K/message icon opens chat. Record access and entry route.
Voice	Click microphone. First-person path auto-posts a successful result; third-person path can put text into a draft for SEND. Current configuration readiness is unverified/blocked.

Task 1 allows 90 seconds; Task 2 allows 120 seconds including the 90-second round; Task 3 allows 90 seconds. Cases start with 50 candles, empty hands, closed menus and first-person view. Initial energy leaves capacity for a three-segment reward. Task 3 starts stationary. The exact initial energy, task traces and assistance are retained in A2-A6.

Appendix B IP2b Headset Testing Plan

B1 Goals tasks assistance and recording structure

The IP2b plan targets a Meta Quest headset with two tracked controllers. It asks whether first-time users can obtain and use rewards, manage energy and scoring during flight, and send readable typed and spoken messages. The plan recruits at least five consenting adult classmates, records prior VR and Sky experience, and uses observation, light think-aloud, 15 ratings and a short interview.

Task	Limit	Required outcomes
T1 Ancestor and items	90 s	Open exchange; obtain reward; deliberately drop and retrieve; consume; aim and throw pottery. A target hit is recorded but is not necessary for core success.
T2 Flight	120 s	Take off, boost, collect a flame and explain score and normal end refill. The round lasts 90 s. View switching is separately recorded and may be skipped for comfort.
T3 Communication	90 s	Place/access table; enter and send "we can fly" by typing and speech; locate and read both results. One retry is allowed per method.

The local target is at least four independent completions out of five for each core task. After 10 seconds without progress, H0 repeats the goal. After a further 10 seconds, H1 points to a visible cue; after another 10 seconds, H2 supplies the control. H1/H2 make a completion assisted. Log substep times, errors, prompts and exact message outputs; technical blocks remain distinct from user errors.

P01, P03 and P05 are assigned typing first; P02 and P04 voice first. This is an uneven descriptive comparison, not a controlled speed experiment. The five-minute task block sits inside a roughly 10-13-minute appointment including preparation and follow-up. Avoid questions during demanding flight or active speech recognition. Ask for local 0-10 comfort ratings and stop immediately for discomfort or a request to stop.

The procedure specifies build and microphone checks, common starting states, energy headroom, closed menus and an empty hand. The case values in Appendix A remain attached to its desktop adaptation.

B2 Desktop and Meta Quest control mapping

Action	Desktop route in Appendix A	Meta Quest route in the plan
Movement and view	WASD and mouse; 1 or 3 changes view.	Left thumbstick moves the player; B changes view.
Exchange	F opens the tree; number keys and Enter select and acquire. Mouse drag is an alternative.	Left-controller X opens the tree. Point with the right ray and hold the right trigger, or pull the icon outward.
Drop and retrieve	Z drops or picks up the item.	A short right-trigger press and release drops the held item; point and press to pick up.
Aim and throw	Hold Q to preview; release to launch.	Hold the right trigger away from the face to preview; release to throw.
Use an edible reward	E consumes the reward.	Bring the virtual reward near the face and hold the right trigger; do not touch the headset with a controller.
Take off and boost	Space takes off or continues ascent; Left Ctrl descends.	Right-controller A takes off or boosts. Each successful boost costs one energy segment.
Typed message	Physical typing; Enter or the on-screen SEND control.	Use the scene keyboard and select SEND with controller interaction.
Spoken message	Click the microphone; unavailable in the Appendix A examples.	Hold the left trigger for English speech input, then check the displayed result.

The platforms share task goals but use different inputs and viewing conditions. A desktop Q-key error is not a Quest trigger error. A physical-keyboard result is not virtual-keyboard performance. Desktop discomfort and view preferences cannot establish headset comfort. Headset conclusions require corresponding headset session records.

Appendix C Speech Input Scenarios

C1 Material status task and timing definition

Activate the microphone, say "we can fly", wait for the result and read it back. One retry is allowed. Waiting runs from the end of speech to the final recognised text and excludes recording time. Completion requires both a matching phrase and reliable readback. Correct but unreadable output is partial; incorrect meaning is not completed.

ID	First wait	First text	Retry wait and text	Readback	Outcome
P01	3.0 s	we can fly	No retry	Correct after looking closely	Completed on first attempt
P02	3.2 s	we can find	3.0 s; we can fly	Correct after retry	Completed after retry
P03	2.9 s	we can try	3.1 s; we cannot fly	Displayed words readable; target meaning incorrect	Not completed
P04	3.1 s	we can fly	No retry	Not reliably readable at normal position	Partially completed
P05	3.0 s	we can fight	3.2 s; we can fly	Correct after retry and close checking	Completed after retry

C2 P01 Waiting and reading effort

Action trace: The first target phrase produces "we can fly" after 3.0 seconds. There is no second microphone activation. The text is read correctly after moving closer to the screen.

Issue: The pause creates uncertainty about whether the microphone captured the speech. Once text appears, the small caption requires extra effort to check. Recognition is correct, but waiting and reading still add effort.

Outcome: Completed on the first attempt. Retain the waiting and readability issues rather than treating completion as an absence of difficulty.

C3 P02 Waiting and an incorrect final word

Action trace: The first result is "we can find" after 3.2 seconds. One retry produces "we can fly" after 3.0 seconds, followed by a correct readback.

Issue: The result must be checked before the incorrect final word can be noticed. Repeating the phrase adds another recording and processing step. Caption size is not the main issue in this case.

Outcome: Completed after retrying. Keep both outputs and both waits; a corrected final message does not erase the first error.

C4 P03 Two outputs that do not match the target

Action trace: The first result is "we can try" after 2.9 seconds. The retry produces "we cannot fly" after 3.1 seconds. Both displayed results can be read, but neither matches "we can fly".

Issue: The second output changes a positive statement into a negative one. This case concerns recognition reliability rather than a failure to display text or a caption-reading problem.

Outcome: Not completed. Speech was attempted and output appeared, so this is neither an unattempted step nor a technical block.

C5 P04 Correct recognition but unreliable readback

Action trace: The correct result "we can fly" appears after 3.1 seconds. The short wait is accepted. The caption is too small to read reliably from the normal viewing position, and there is no speech retry.

Issue: The output matches the target, but checking it remains difficult. Correct recognition and readable presentation are separate requirements.

Outcome: Partially completed. Keep the reading difficulty separate from recognition accuracy.

C6 P05 Waiting recognition and reading effort

Action trace: The first result is "we can fight" after 3.0 seconds. Looking closer reveals the incorrect word. A retry produces "we can fly" after 3.2 seconds, followed by another careful check and a correct readback.

Issue: Small captions delay error detection. Correcting the error then requires another spoken attempt and another wait. The three issues combine to add steps to a short message.

Outcome: Completed after retrying, with reading difficulty retained. This does not demonstrate reliable first-attempt recognition.

Appendix D Concept Development and Design Principles

D1 Original XR concept and proposed interactions

The original concept selected pure VR to preserve Sky's continuous artistic world. It proposed head-mounted viewing, two tracked controllers and a finger-worn input device. Its three goals were open-sky exploration, direct social/object interaction and multimodal communication.

Area	Original proposal	Proposed benefit considered in this evaluation
Flight	Downward hand motion for take-off, arms spread for gliding, first/third-person views.	Physical gestures and view choice would feel natural while maintaining orientation and comfort.
Ancestor and items	Spatial exchange trees, direct item handling, overhead menu, social prop placement.	Spatial cues and body actions would reduce reliance on conventional menus.
Communication	Speech recognition and quiet finger-worn air-typing.	Different input methods would support both active play and quiet settings.

D2 Low-fidelity models

The concept report includes sketches over screenshots and paper models for viewpoint switching, speech display, ancestor-tree interaction, an immersive top menu and prop placement. They represent positions and action sequences, not functioning tracking, recognition or measured fatigue.

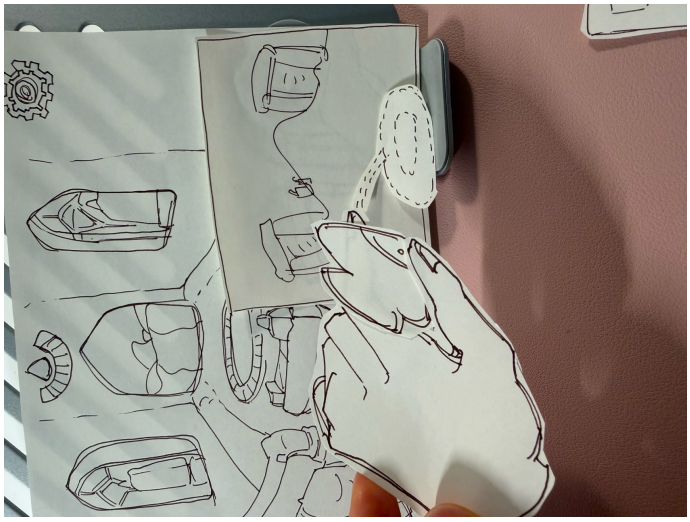


Figure D1. Original paper model labelled as the immersive top-menu interaction in the concept report.

D3 Week 6 five design principles

Text Readability: make essential text easy to find and read. Ergonomics: consider reach, posture and repeated movement. Brightness Changes: consider how changes in light affect attention and viewing. Avoiding Simulator Sickness: consider movement and viewpoint comfort. Button Placement: make controls accessible and their purpose clear.

Appendix E IP1 Testing Plan and Interview Evidence

E1 Plan and prototype conditions

IP1 ran in Unity with keyboard and mouse. The plan used task-based testing, observation, audio recording and semi-structured interviews. It addressed the ancestor tree, obtaining/carrying/wearing/throwing a mask, discovering the T-key keyboard and expectations about future controller communication. It proposed recording errors, repeated attempts and independent discovery.

The prototype used E for mask interaction, Q/P for throwing-related actions and T for the keyboard. The records describe an incomplete ancestor extraction process. As a result, the intended continuous sequence could not be treated as fully demonstrated in IP1.

E2 Selected interview observations

Record	Recorded observation	Representative wording in the source
Interview 1	The evaluator explained E and the Q/P/T controls. Wearing followed guidance; the keyboard was introduced verbally.	"At the beginning, I didn't know what the keys did, I only knew WASD."
Interview 2	Two years of Sky experience; pickup key still needed explanation; camera adjusted repeatedly.	"Which key is for picking up?" and "Rotating the camera is still a bit troublesome."

Record	Recorded observation	Representative wording in the source
Interview 3	Q and P needed explanation; object interaction was enjoyable after learning.	"Q is just throwing it out randomly; P requires a long press to select a position."
Interview 4	Unnatural movement and camera positioning; limited entries for tree and keyboard tasks.	"Moving around is very weird."
Interview 5	Sparse individual notes; NPC overlap restricted visibility and mouse input was needed for rotation.	"Vision is limited, making it easy to overlap with NPCs."

E3 Recording limits relevant to the reflection

The record provides useful descriptions and interview statements but no consistent complete set of task timings, error counts or help levels. Missing tree/keyboard entries for later interviews remain missing, not successful outcomes. Initial discovery, guided completion and opinions about future controller typing must be distinguished. Familiarity with Sky did not remove the new control and camera difficulties in Interview 2.

Appendix F IP2a Testing Plan and Results

F1 Focused task structure and decision rules

The focused plan addresses ancestor discovery, mask acquisition and wearing. Formal XR testing uses Meta Quest; desktop trial runs are separate. Basic input practice uses a neutral object before timed tasks, without demonstrating the target mask or tree sequence.

Task	Limit	Starting state and success endpoint
T1 Open interface	40 s	Ancestor visible, tree closed; success means the correct interface opens.
T2 Obtain mask	60 s	Tree open, mask available, empty hand; the correct mask is generated and held, not merely selected.
T3 Wear mask	50 s	Mask held; the system equips it. Record whether the user's completion judgement matches the actual state.

At least four of five independently completing each task is the local target. H0 means no help; H1 is a general cue, H2 an exact action and H3 a demonstration. If a previous task fails, I may prepare the next starting state but must record this separately; it does not prove an independently completed chain. The plan separates task outcomes, state understanding, errors, camera events and short follow-up questions.

F2 Individual ratings and means

Measure	P01	P02	P03	P04	P05	Mean / 5
T1 Interface discovery	5	3	4	3	5	4.0
T1 Purpose clarity	5	4	5	4	5	4.6
T2 Obtaining the mask	4	3	4	4	5	4.0
T2 Obtaining feedback	4	3	5	4	4	4.0
T3 Wearing the mask	5	4	5	4	5	4.6
T3 Equip feedback	5	4	5	3	5	4.4
Camera and movement	4	3	4	5	4	4.0
On-screen guidance	4	3	5	4	4	4.0
Independent confidence	3	4	5	4	5	4.2
Overall satisfaction	5	4	5	5	5	4.8

The supplied IP2a evaluation identifies three participants as uncertain about selection versus acquisition. Appearance changes supported recognition of the equipped state across the five records. These descriptive ratings do not by themselves establish independent task completion.

F3 Written feedback on acquisition wearing and camera behaviour

P01

What indicated that the mask had been obtained and equipped? The mask disappeared from the tree and appeared on my character. I also noticed a short sound and a visual change, so I knew it was equipped.

Which step caused uncertainty, and what was expected? I was only a little unsure when taking the mask from the tree. I expected it to move directly into my hand.

Which part worked best, and why? Putting on the mask worked best. The action was simple and the result was easy to see.

What should still be changed, and why? I would make the feedback after obtaining the mask slightly stronger, because the change was quite quick.

P02

What indicated that the mask had been obtained and equipped? The mask was no longer on the tree, and later I could see it on the character. This helped me understand that the action was successful.

Which step caused uncertainty, and what was expected? Obtaining the mask was the most uncertain step. I was not sure if I needed to click, hold or drag it. I expected a clearer instruction.

Which part worked best, and why? The highlighting of interactive objects was helpful. It showed me which object I could use.

What should still be changed, and why? I would add a short instruction before taking the mask. This would help new players who do not know the controls.

P03

What indicated that the mask had been obtained and equipped? I saw the mask move from the tree and the interface changed. When I equipped it, the character model changed and I heard a confirmation sound.

Which step caused uncertainty, and what was expected? There was no major uncertain step. The visual guidance made the process understandable.

Which part worked best, and why? The system feedback worked best. I could understand when each action was completed without asking the facilitator.

What should still be changed, and why? I would reduce the small delay after selecting the mask. The interaction would feel more responsive.

P04

What indicated that the mask had been obtained and equipped? The mask appeared on the character’s face. The visual change was clear, so I knew I had finished the task.

Which step caused uncertainty, and what was expected? I was slightly uncertain when standing close to the ancestor because the camera angle changed. I expected the camera to stay behind the character.

Which part worked best, and why? Equipping the mask was the clearest part. It only required one simple action and gave clear feedback.

What should still be changed, and why? I would make the camera automatically face the important object. This could make the interaction more comfortable.

P05

What indicated that the mask had been obtained and equipped? The item changed position after I selected it. When I put it on, I could see the mask on the character and the interface showed that it was equipped.

Which step caused uncertainty, and what was expected? For a short moment, I was not sure whether the mask was only selected or already obtained. I expected a clearer status message.

Which part worked best, and why? The complete task flow worked well. The three tasks felt connected and the controls were consistent.

What should still be changed, and why? I would add a more visible confirmation after obtaining the mask. This would make the item state easier to understand.

Appendix G IP2b Questionnaire and Follow-up Prompts

G1 Questionnaire instructions and fifteen rating items

Tell us about your experience

After the desktop session - 3-4 min

Answer only from what you experienced. Choose N/A if you did not experience an item. This is a custom diagnostic questionnaire, not a validated usability or sickness scale. In Q8, boost means desktop ascent. If voice was unavailable and untried, select N/A for Q14; rate an actual failed attempt from experience.

Participant ID: _____	Actual test date and time: _____
Actual input order: _____	Completion context: _____

Date and time: yyyy/mm/dd, --:--. Australia/Brisbane (UTC+10). Do not enter the survey-entry time here.

[ ] I agree to provide anonymous feedback for this course project. I can skip an item using N/A and stop at any time.

Scale: 1 Strongly disagree; 2 Disagree; 3 Neutral; 4 Agree; 5 Strongly agree; N/A Not experienced.

No.	Statement	Select one response
01	I noticed the ancestor when I approached the monument.	1 / 2 / 3 / 4 / 5 / N/A
02	I could work out how to open the exchange tree.	1 / 2 / 3 / 4 / 5 / N/A
03	I could obtain the reward I intended to select.	1 / 2 / 3 / 4 / 5 / N/A
04	I could distinguish dropping an item from throwing it.	1 / 2 / 3 / 4 / 5 / N/A
05	The throwing arc and landing circle helped me aim.	1 / 2 / 3 / 4 / 5 / N/A
06	I understood how to consume the energy reward.	1 / 2 / 3 / 4 / 5 / N/A
07	I understood the resource change after consuming the reward.	1 / 2 / 3 / 4 / 5 / N/A
08	I understood how to take off and boost.	1 / 2 / 3 / 4 / 5 / N/A
09	I could tell how much flight energy remained.	1 / 2 / 3 / 4 / 5 / N/A
10	I understood how collecting flames affected my score.	1 / 2 / 3 / 4 / 5 / N/A
11	I understood why my energy refilled when the round ended.	1 / 2 / 3 / 4 / 5 / N/A
12	I remained oriented when switching views.	1 / 2 / 3 / 4 / 5 / N/A
13	I could enter and send the typed message.	1 / 2 / 3 / 4 / 5 / N/A
14	I could send the intended message using my voice.	1 / 2 / 3 / 4 / 5 / N/A
15	I could read the instructions, resource displays and chat bubbles.	1 / 2 / 3 / 4 / 5 / N/A

G2 Comfort and preferences

Final discomfort: _____

0 = none, 10 = extreme. This is a local check, not a clinical scale.

Preferred view: _____

Preferred input: _____

Which moment was most confusing, and what did you expect?

Describe the moment, or write 'none'.

Which single change would help you most?

Optional: Why did you prefer that view or input method?